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MAE-Optimized Machine Learning Ensemble for Post-Processing GFS Solar Irradiance Forecasts in East Asia

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Post-processing GFS on the absolute error instead of the squared error

In contrast to typical post-processing, the ensemble is trained on the absolute error instead of the squared error. Across three climates in East Asia, it reduces the error of raw GFS by up to 20%, with the lowest nMAE at every station.

11–18% lower nMAE than raw GFS

13–20% lower nRMSE than raw GFS

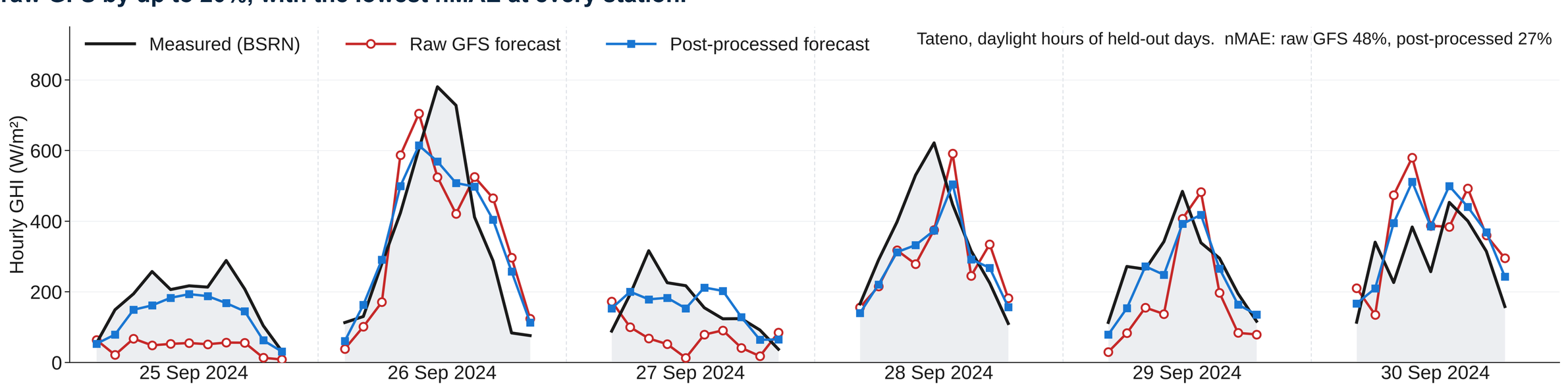
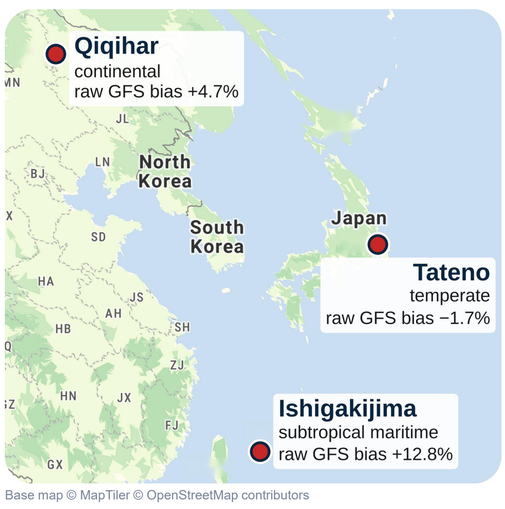


Fig. 1. On 25 and 27 September GFS forecasts much thicker cloud than was observed; the calibration removes most of that error. Held-out days at Tateno, chosen for illustration; scores for the whole year are in Fig. 3.

Context

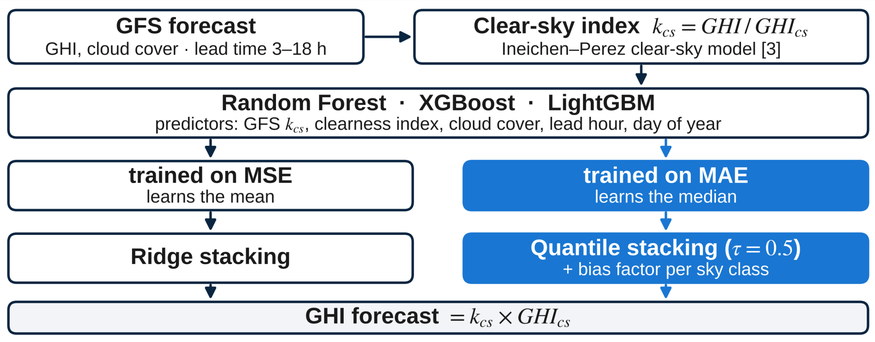


- GFS forecasts are free and global, but show bias at all three chosen stations (map).
- Machine-learning post-processing against ground data reduces these errors [1].
- The loss function decides what the model learns: the squared error leads to the mean, the absolute error to the median [2].

Objective

Is an ensemble trained on the absolute error, from the base learners to the stacking, a better calibration of GFS than the traditional MSE-trained one?

Methodology



Data: BSRN 1-min GHI [4], 2024, hourly, Sun above 5°. **Reference:** MOS-MLR.

Test: 5-fold cross-validation. Each month is split into five blocks of about 6 days; fold *k* holds out block *k* of every month.

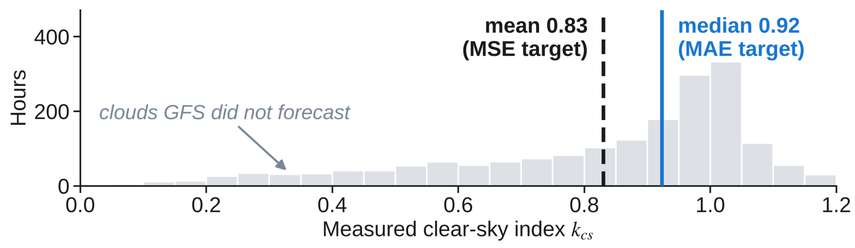


Fig. 2. When GFS forecasts a clear sky, one hour in four is in fact cloudy. These hours (k_{cs} < 0.7) split mean and median. Ishigakijima, forecasted $k_{cs,GFS}$ > 0.9, measured k_{cs} ≤ 1.2.

Results and Discussion

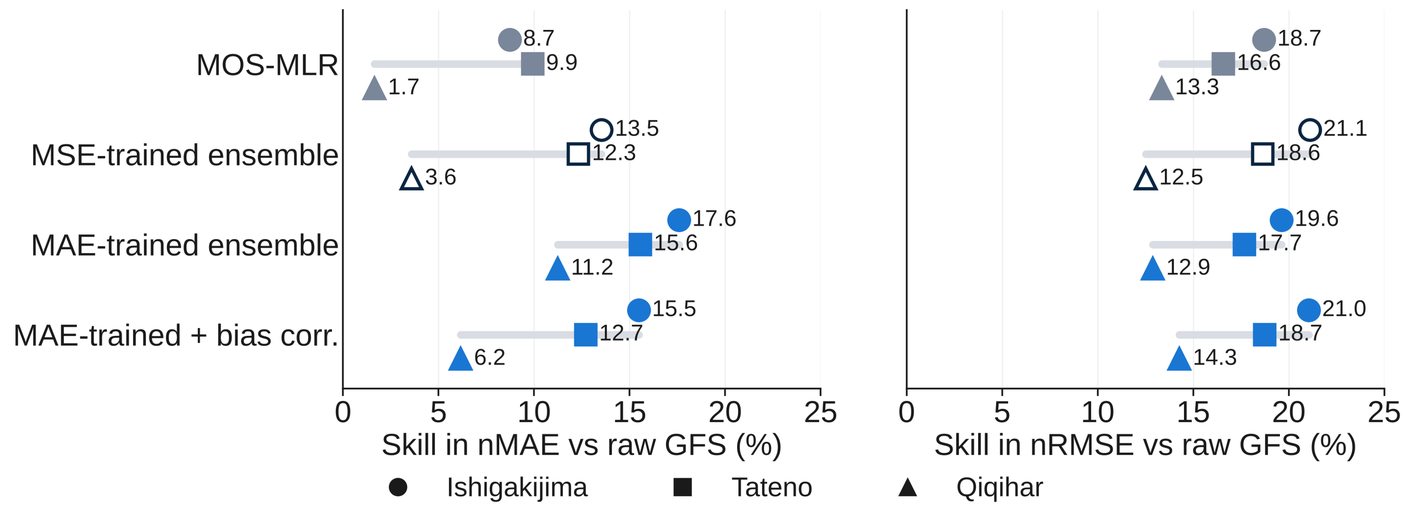


Fig. 3. MAE training gives the highest nMAE skill at every station; in nRMSE the ensembles are practically equivalent. Skill against raw GFS, all 2024 daylight hours, 5-fold blocked test.

nMAE (%)
nRMSE (%) below
LightGBM (MAE)

Trained at \ Tested at	Ishigakijima	Tateno	Qiqihar
Ishigakijima	**25.8** 36.1	26.4 37.0	25.8 39.1
Tateno	29.4 43.4	**20.6** 32.4	19.3 31.5
Qiqihar	30.6 46.7	23.7 36.7	**19.5** 33.7
all three, pooled	28.2 41.9	20.9 33.3	18.9 31.4
pooled + station ID	**26.5** 38.4	**20.9** 33.6	**18.8** 31.9
raw GFS	**32.7** 48.8	**26.9** 43.3	**20.5** 36.0

Fig. 4. The calibration is climate-specific: trained in the subtropics, it is worse than raw GFS at Qiqihar. A pooled model with a station identifier matches the local ones.

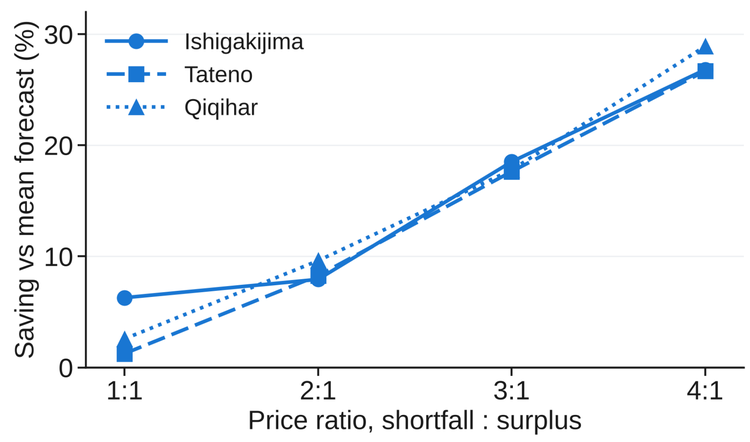


Fig. 5. The more asymmetric the imbalance price, the more the quantile forecast saves. Quantile τ = 1/(1+r) instead of the mean, for a shortfall priced r times a surplus.

Conclusions

1. Post-processing in clear-sky index space reduces the error of raw GFS at all three stations: nMAE by 11–18%, nRMSE by 13–20%.
2. Training on the absolute error lowers nMAE at every station; with a sky-class bias correction, nRMSE and bias also match the MSE-trained ensemble.
3. One pooled model with a station identifier is almost as accurate as the local models. With asymmetric imbalance prices, the matching quantile cuts the cost by up to 30%.

Limits: one year of data, three stations, lead time 3–18 h.

Acknowledgment

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References

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